



# **DEPARTMENT OF COMPUTER SCIENCE**

## **BCS 1 (A)**

### **FINAL PROJECT OF PROGRAMMING FUNDAMENTALS**

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# **PROJECT TOPIC: HOTEL RESERVATION SYSTEM**

## **PROGRAM:**

## **SOURCE CODE:**

// ConsoleApplication21.cpp : This file contains the 'main' function. Program execution begins and ends there.

//

#include <stdio.h>

#include <stdbool.h>

#include <stdlib.h>

#define numRooms 50

bool rooms[numRooms] = { false };

int countAvailableRooms() {

    int count = 0;

    for (int i = 0; i < numRooms; ++i) {

        if (!rooms[i]) {

            count++;

        }

    }

    return count;

}

void displayMenu() {

    printf("\nHotel Reservation System\n");

    printf("1. Book a room\n");

    printf("2. Check room availability\n");

    printf("3. Display booked rooms\n");

    printf("4. Cancel reservation\n");

    printf("5. Display available rooms\n");

    printf("6. Display room counts\n");

    printf("7. Display room details\n");

```
printf("8. Modify reservation\n");

printf("9. Generate reservation report\n");

printf("10. Exit\n");
}

void bookRoom() {

    int roomNumber;

    printf("Enter room number to book: ");

    scanf_s("%d", &roomNumber);

    if (roomNumber >= 1 && roomNumber <= numRooms) {

        if (rooms[roomNumber - 1]) {

            printf("Room already booked. Try another room.\n");

        }

        else {

            rooms[roomNumber - 1] = true;

            printf("Room %d booked successfully.\n", roomNumber);

        }

    }

    else {

        printf("Invalid room number. Please enter a number between 1 and %d.\n", numRooms);

    }

}

void cancelReservation() {

    int roomNumber;

    printf("Enter room number to cancel reservation: ");

    scanf_s("%d", &roomNumber);

    if (roomNumber >= 1 && roomNumber <= numRooms) {

        if (rooms[roomNumber - 1]) {

            rooms[roomNumber - 1] = false;

            printf("Reservation for Room %d canceled successfully.\n", roomNumber);

        }

    }

}
```

```

    }
    else {
        printf("No reservation found for Room %d.\n", roomNumber);
    }
}
else {
    printf("Invalid room number. Please enter a number between 1 and %d.\n", numRooms);
}
}

void displayAvailableRooms() {
    printf("\nAvailable Rooms:\n");
    for (int i = 0; i < numRooms; ++i) {
        if (!rooms[i]) {
            printf("Room %2d - Available\n", i + 1);
        }
    }
    printf("\n");
}

void displayRoomCounts() {
    int availableRooms = countAvailableRooms();
    printf("Total Rooms: %d\n", numRooms);
    printf("Available Rooms: %d\n", availableRooms);
    printf("\n");
}

void displayRoomDetails() {
    int roomNumber;
    printf("Enter room number to display details: ");
    scanf_s("%d", &roomNumber);
}

```

```
if (roomNumber >= 1 && roomNumber <= numRooms) {  
    printf("\nRoom %d Details:\n", roomNumber);  
    printf("Capacity: 2 persons\n");  
    printf("services: TV, Wi-Fi, Bathroom\n");  
    printf("Price per night: $100\n");  
}  
else {  
    printf("Invalid room number. Please enter a number between 1 and %d.\n", numRooms);  
}  
}  
  
void modifyReservation() {  
    int roomNumber;  
    printf("Enter room number to modify reservation: ");  
    scanf_s("%d", &roomNumber);  
    if (roomNumber >= 1 && roomNumber <= numRooms) {  
        if (rooms[roomNumber - 1]) {  
            printf("Reservation for Room %d modified successfully.\n", roomNumber);  
        }  
        else {  
            printf("No reservation found for Room %d.\n", roomNumber);  
        }  
    }  
    else {  
        printf("Invalid room number. Please enter a number between 1 and %d.\n", numRooms);  
    }  
}  
  
void generateReservationReport() {  
    printf("\nReservation Report:\n");
```

```
    for (int i = 0; i < numRooms; ++i) {  
        if (rooms[i]) {  
            printf("Room %2d - Reserved\n", i + 1);  
        }  
    }  
    printf("\n");  
}  
  
int main() {  
    int choice;  
    do {  
        displayMenu();  
        printf("Enter your choice: ");  
        scanf_s("%d", &choice);  
        switch (choice) {  
            case 1:  
                bookRoom();  
                break;  
            case 2:  
                displayAvailableRooms();  
                break;  
            case 3:  
                displayRoomCounts();  
                break;  
            case 4:  
                cancelReservation();  
                break;  
            case 5:  
                displayAvailableRooms();  
                break;
```

```
case 6:
    displayRoomCounts();
    break;
case 7:
    displayRoomDetails();
    break;
case 8:
    modifyReservation();
    break;
case 9:
    generateReservationReport();
break;
case 10:
    printf("Exiting the program.\n");
    break;
default:
    printf("Invalid choice. Please enter a valid option.\n");
}
} while (choice != 10);
return 0;
}
```

## OUTPUT:

 Microsoft Visual Studio Debug Console

```
Hotel Reservation System
1. Book a room
2. Check room availability
3. Display booked rooms
4. Cancel reservation
5. Display available rooms
6. Display room counts
7. Display room details
8. Modify reservation
9. Generate reservation report
10. Exit
Enter your choice: 1
Enter room number to book: 45
Room 45 booked successfully.
```

## Hotel Reservation System

1. Book a room
2. Check room availability
3. Display booked rooms
4. Cancel reservation
5. Display available rooms
6. Display room counts
7. Display room details
8. Modify reservation
9. Generate reservation report
10. Exit

Enter your choice: 2

### Available Rooms:

Room 1 - Available  
Room 2 - Available  
Room 3 - Available  
Room 4 - Available  
Room 5 - Available  
Room 6 - Available  
Room 7 - Available  
Room 8 - Available  
Room 9 - Available  
Room 10 - Available  
Room 11 - Available  
Room 12 - Available  
Room 13 - Available  
Room 14 - Available  
Room 15 - Available  
Room 16 - Available  
Room 17 - Available  
Room 18 - Available



Room 18 - Available  
Room 19 - Available  
Room 20 - Available  
Room 21 - Available  
Room 22 - Available  
Room 23 - Available  
Room 24 - Available  
Room 25 - Available  
Room 26 - Available  
Room 27 - Available  
Room 28 - Available  
Room 29 - Available  
Room 30 - Available  
Room 31 - Available  
Room 32 - Available  
Room 33 - Available  
Room 34 - Available  
Room 35 - Available  
Room 36 - Available  
Room 37 - Available  
Room 38 - Available  
Room 39 - Available  
Room 40 - Available  
Room 41 - Available  
Room 42 - Available  
Room 43 - Available  
Room 44 - Available  
Room 46 - Available  
Room 47 - Available  
Room 48 - Available

Room 49 - Available

Room 50 - Available

Hotel Reservation System

1. Book a room
2. Check room availability
3. Display booked rooms
4. Cancel reservation
5. Display available rooms
6. Display room counts
7. Display room details
8. Modify reservation
9. Generate reservation report
10. Exit

Enter your choice: 6

Total Rooms: 50

Available Rooms: 50

Hotel Reservation System

1. Book a room
2. Check room availability
3. Display booked rooms
4. Cancel reservation
5. Display available rooms
6. Display room counts
7. Display room details
8. Modify reservation
9. Generate reservation report
10. Exit

Enter your choice: 7

```
Enter your choice: 7
Enter room number to display details: 43

Room 43 Details:
Capacity: 2 persons
services: TV, Wi-Fi, Bathroom
Price per night: $100

Hotel Reservation System
1. Book a room
2. Check room availability
3. Display booked rooms
4. Cancel reservation
5. Display available rooms
6. Display room counts
7. Display room details
8. Modify reservation
9. Generate reservation report
10. Exit
Enter your choice: 8
Enter room number to modify reservation: 45
No reservation found for Room 45.

Hotel Reservation System
1. Book a room
2. Check room availability
3. Display booked rooms
4. Cancel reservation
5. Display available rooms
6. Display room counts
```

Select Microsoft Visual Studio Debug Console

```
1. Book a room
2. Check room availability
3. Display booked rooms
4. Cancel reservation
5. Display available rooms
6. Display room counts
7. Display room details
8. Modify reservation
9. Generate reservation report
10. Exit
Enter your choice: 9

Reservation Report:

Hotel Reservation System
1. Book a room
2. Check room availability
3. Display booked rooms
4. Cancel reservation
5. Display available rooms
6. Display room counts
7. Display room details
8. Modify reservation
9. Generate reservation report
10. Exit
Enter your choice: 10
Exiting the program.

C:\Users\zeshan\source\repos\ConsoleApplication21\x64\Debug\Cor
Press any key to close this window . . .
```

